

PROJECT NIGHTWATCH

The Unattended Autonomy and Overnight Operations System

Governing Architecture, Operational Safety, and Product Requirements Document



Governing Principle

An unattended Codex session must remain capable of productive work across the authorized night, but no single objective may monopolize that session through repeated failure, candidate churn, uncontrolled scope growth, or unapproved cost.

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Prepared for the Voyagewright / Chronicles platform

Governing Architecture, Operational Safety, and Product Requirements Document

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Governing Principle An unattended Codex session must remain capable of productive, useful work across the authorized night, but no single objective may monopolize that session through repeated failures, candidate churn, uncontrolled scope growth, or unapproved cost. Nightwatch contains a pathological objective, preserves its evidence, and redirects the remaining capacity to the next safe task.

Field	Value
Program	Project Nightwatch
Subsystem	The Unattended Autonomy and Overnight Operations System
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Status	Governing baseline candidate
Primary operating surfaces	Codex sessions, supervised unattended runners, GitHub, Sounding Line, Project Trim, Breakwater, Bridgewatch
Core implementation rule	Continue productive unattended work across a governed queue; park pathological objectives without ending the useful night
Founding incident	INC - T1 - 2026 - 08 - 19 - RUNAWAY - ITERATION - 001

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1. Executive Summary

Project Nightwatch establishes a governed system for long-running, unattended Codex work. Its purpose is not to shorten every unattended session, prohibit overnight work, or require the owner to babysit automation. Its purpose is to make unattended work substantially more productive by ensuring that the entire authorized night cannot be consumed by one pathological objective.

Nightwatch separates the budget of the overall unattended session from the budget of each objective inside that session. A Nightwatch session may be authorized to work for eight hours, while an individual objective may receive only 90 minutes, two repeated failure signatures, two candidate refreezes, two prerequisite repairs, and five total pull requests. When an objective consumes its local budget or stops producing material progress, Nightwatch preserves its exact state, marks it parked, and continues with the next authorized objective. The night continues; the loop does not.

The founding incident demonstrated why this separation is necessary. An unattended browser-contract and Tideglass cleanup objective entered a repeated repair-and-refreeze cycle. The objective lineage created 33 pull requests from #302 through #334, merged six supporting repairs, closed 23 attempts without merge, left four successors open, grew the surviving candidate to 32 files, exhausted the operator-declared remaining weekly quota, and crossed into separately purchased usage without completing the primary product merge. The final surviving candidate failed protected binding at `Locate sealed explicit authority`. Repository activity then stopped while the Codex session continued consuming resources locally.

Nightwatch converts this failure history into permanent system behavior. It introduces a Night Plan, a prioritized work queue, global and objective budgets, material-progress detection, semantic failure fingerprints, lane-level circuit breakers, one-active-candidate rules, scope-growth tripwires, persistent restart-safe ledgers, credit-aware scheduling, execution leases, Bridgewatch telemetry, and morning handoff records.

Nightwatch does not replace Sounding Line, Project Trim, Breakwater, Bridgewatch, or Deepwater. It composes their accepted contracts:

- Sounding Line remains the authority for verification planning, evidence validity, execution, and release decisions.

- Project Trim remains the authority for Codex context and inference efficiency.
- Breakwater remains the owner of deployment, infrastructure, release consumption, and platform operations.
- Bridgewatch remains the operator-facing observation and command-center surface.
- Deepwater remains the audit, ownership, findings, and realization-evidence system.

Core Product Outcome

The owner may authorize meaningful work before going to sleep and return to a morning record showing completed objectives, parked blockers, exact evidence, bounded resource use, and a repository that advanced without allowing any single task to become an eight-hour credit furnace with a pull-request breeding program.

2. Project Identity and Vision

2.1 Program Name

The subsystem is named Project Nightwatch.

A night watch is a deliberate period of responsibility while others sleep. It does not imply inactivity, fear of operation, or a timer waiting to ring. A good watch maintains progress, observes changing conditions, records what matters, responds within authority, and leaves a trustworthy handoff for the next watch.

The repository contains no prior Nightwatch project identity at the reviewed baseline. The name is distinct from:

- Breakwater, which owns deployment, infrastructure, release consumption, and platform operations.
- Bridgewatch, which provides observation and command-center presentation.
- Watchglass, which owns recognition and external verification-provider truth.

2.2 Formal Subtitle

The Unattended Autonomy and Overnight Operations System

2.3 Vision

Nightwatch should allow Codex to work more effectively when the owner is absent than an ordinary interactive chat can work while being constantly steered. The absence of the owner should trigger better planning, stronger state preservation, bounded discretion, and productive task switching - not timid inactivity and not infinite persistence on one blocked objective.

Nightwatch treats the night as a portfolio of authorized work. It optimizes useful completed outcomes across the session rather than maximizing effort spent on the first task. It may pursue a difficult objective, diagnose failures, land bounded prerequisites, and requalify a candidate. It may not mistake the theoretical existence of another possible attempt for unlimited authorization.

2.4 Primary Use Cases

- Finish one or more already-defined repository objectives while the owner sleeps.
- Monitor CI and continue independent work while another objective waits.
- Repair a bounded prerequisite, then resume the original candidate once.
- Park a repeatedly failing objective and move to a secondary authorized task.
- Reconcile protected-main movement without recreating an unbounded PR lineage.
- Perform safe, preauthorized housekeeping after primary work completes.
- Preserve a restart-safe execution ledger through context compaction, process restart, or laptop reboot.
- Produce a concise morning report with completed work, parked work, resource use, and exact next actions.

2.5 Governing Interpretation of "Run All Night"

Within Nightwatch, "run all night" means:

Continue working through the authorized Night Plan until the wake-up deadline, global budget, or terminal queue state. Pursue each objective only within its individual limits. When an objective becomes repetitive, expensive, scope-expanding, or blocked, checkpoint and park it, then immediately continue with the next authorized task.

It does not mean:

Continue regenerating the same objective until completion regardless of cost, elapsed time, repeated failure, scope growth, or candidate count.

3. Founding Tier 1 Incident

3.1 Incident Identity

Field	Value
Incident ID	INC-T1-2026-08-19-RUNAWAY-ITERATION-001
Severity	Tier 1
Incident class	Runaway unattended iterative loop
Primary repository	Kgray44/treasurehuntSoT
Objective family	Browser product contracts, Tideglass fixtures, superseded browser-candidate cleanup
Reviewed lineage	PR #302 through PR #334
Final reviewed main	d7e3f2f873139e3f7353bf288124f50543ce2aac

3.2 Intended Overnight Outcome

The unattended request contained three bounded goals:

1. Merge the surviving browser-baseline product-contract candidate.
2. Resolve the Tideglass fixture chain around PRs #296 and #301.
3. Close superseded Lanternwake and browser-contract candidates.

The task was explicitly expected to conserve a nearly exhausted weekly quota. The mandate also instructed the agent to continue until completion through the night. The resulting prompt contained an unsafe contradiction: credit conservation was declared important, but later language said not to stop merely to conserve quota while a viable path remained.

3.3 Observed Outcome

Within the reviewed #302 through #334 lineage:

Outcome	Count
New pull requests	33
Merged	6
Closed without merge	23
Still open at review	4
Primary product-contract objective merged	No

Merged supporting work:

- #305 - partition mixed Tideglass browser fixtures.
- #311 - isolate Shipwright browser fixture.
- #315 - register Shipwright browser engine.
- #318 - normalize browser-selection partitions.
- #330 - admit the governed Shipwright runner.
- #333 - admit generated P34 browser ledgers.

The surviving PR #334 contained:

- 3 commits.
- 32 changed files.
- 1,671 additions.
- 1,089 deletions.
- Product changes, browser baselines, P34 retirement identities, and a Deepwater governance identity update.

Its protected binding failed at:

Locate sealed explicit authority

No successful normal candidate qualification envelope was attached to the reviewed head. GitHub activity ceased after the successor was opened, but unattended computation continued locally.

3.4 Attribution Boundary

The owner reported that this was almost the only active Codex chat during the period, while one or two other chats may have operated briefly. Nightwatch therefore treats the objective-linked PR lineage, failure patterns, and preserved GitHub evidence as authoritative incident data. It does not claim that every unrelated repository event during the same wall-clock period came from one session.

3.5 Why This Is Tier 1

The incident qualifies as Tier 1 because it combined:

- Exhaustion of a declared near-empty weekly quota.
- Unapproved transition into separately purchased usage.
- Hours of unattended repetition.
- Dozens of created and abandoned repository objects.
- Failure to complete the primary requested deliverable.
- Expansion beyond the original task scope.
- Absence of an effective operator alert or circuit breaker.
- Continued effort after repository evidence showed the active lane had already failed.

The severity is based on control-plane failure, not merely expense. The system could not distinguish determined autonomy from pathological persistence.

4. Problem Statement and Root Cause

4.1 Primary Failure Mode

The agent entered this cycle:

```

create product candidate
-> encounter authority or fixture prerequisite
-> create prerequisite repair
-> merge prerequisite
-> refreeze product candidate
-> encounter another prerequisite or identity mismatch
-> create another successor
-> repeat

```

Each local action was plausibly related to completion. The total sequence was unacceptable.

4.2 Root Causes

1. Unbounded completion mandate. Completion language had no externally enforced objective ceiling.
2. Conflicting budget language. The prompt said to conserve quota but also said not to stop for quota while a viable path remained.
3. No distinction between session persistence and objective persistence. The agent treated "keep working" as "keep working on this one lane."
4. No semantic loop detector. Different PR numbers and SHAs obscured equivalent repeated failure classes.
5. No material-progress test. Activity was counted as progress even when it merely regenerated candidates.
6. No candidate-lineage budget. Every replacement was treated as a fresh opportunity.
7. No scope-growth breaker. A narrow product candidate expanded into a 32-file cross-governance envelope.
8. Late authority preflight. Candidates were repeatedly created before the full qualification route was proven.
9. No productive fallback queue. A blocked primary objective had nowhere else to send the remaining night.
10. No global operator telemetry. The owner could not observe the problem while asleep.

4.3 Contributing Conditions

- Protected main advanced repeatedly.
- Sounding Line correctly failed closed on missing or stale authority.
- Anti-self-authorization boundaries required some prerequisites to land separately.
- The agent was authorized to reconcile moving main automatically.
- The repository contained older related PR families whose boundaries were easy to absorb into the overnight objective.

These conditions explain the difficulty. They do not justify unlimited iteration.

5. Non-Negotiable Design Principles

Principle	Requirement
Unattended-First	Nightwatch exists to improve unattended productivity, not prohibit it.
Session Persistence, Objective Boundedness	The night may continue after a single objective is parked.
Productive Work Over Visible Motion	New branches, PRs, logs, and reruns count only when they materially advance the Night Plan.
External Budgets	The working agent cannot silently extend its own limits.
Park and Pivot	A pathological lane is preserved and left; the remaining queue continues.
One Active Candidate	Each deliverable has at most one active product candidate.
Semantic Loop Detection	Equivalent failures are recognized across changing wording, SHAs, and PR numbers.

Principle	Requirement
Evidence Preservation	Parking never discards useful commits, logs, receipts, or diagnosis.
Credit Stewardship	Project Trim cost and context signals influence scheduling and repetition decisions.
Sounding Line Authority	Nightwatch never weakens, bypasses, or fabricates release evidence.
Restart Safety	Ledger, budgets, and counters survive compaction and restart.
Honest Morning Handoff	The owner receives completed outcomes, exact blockers, and resource use without narrative camouflage.

Hard Boundary Nightwatch may choose which authorized objective to run next, but it may not reinterpret a failed Sounding Line decision as permission to bypass qualification, lower verification, modify protection, or manufacture authority.

6. Scope and Non-Goals

6.1 Governed Scope

- Night Plan creation and validation.
- Prioritized objective queues.
- Global and per-objective budgets.
- Failure fingerprinting and iterative-loop detection.
- Material-progress measurement.
- Candidate and PR lineage accounting.
- Scope-growth monitoring.
- Park, pivot, resume, complete, and abort transitions.
- Persistent unattended ledgers.
- Restart and morning-handoff behavior.
- Credit-aware scheduling through Project Trim contracts.
- Execution leases and mutation accounting.
- Bridgework visibility and alerts.
- Tier 1 incident capture and replay testing.

6.2 Non-Goals

- Nightwatch is not a replacement for Codex.
- Nightwatch is not a replacement for Sounding Line qualification.
- Nightwatch is not a new GitHub API or branch-protection authority.
- Nightwatch does not guarantee that every overnight objective completes.
- Nightwatch does not force the entire night to stop when one task becomes blocked.
- Nightwatch does not treat all long-running work as runaway work.
- Nightwatch does not permit unbounded paid usage merely because the owner requested overnight work.
- Nightwatch does not automatically absorb unrelated repository issues into the current objective.

- Nightwatch does not hide partial progress or call prerequisite activity equivalent to completion of the requested deliverable.

7. Canonical Ownership and Cross-Project Boundaries

7.1 Nightwatch Contract

Nightwatch owns:

Unattended Codex orchestration, persistent overnight execution, objective budgeting, loop containment, productive task switching, and morning handoff.

7.2 Existing Project Contracts

Project	Existing contract	Nightwatch relationship
Sounding Line	Verification planning, evidence validity, execution, and release decisions	Consumes qualification state; never replaces authority
Project Trim	Codex context and inference efficiency	Consumes cost, capsule, reuse, and context-efficiency signals
Breakwater	Deployment, infrastructure, release consumption, and platform operations	Uses operational runtime, process supervision, and platform facilities
Bridgewatch	Read-only project and repository command-center visibility	Publishes Nightwatch state, budgets, progress, and alerts
Deepwater	Capability audit, ownership, findings, and realization evidence	Registers ownership and audits implementation completeness
Fairlead	GitHub interaction and quota control plane	Consumes bounded GitHub operation and quota signals where applicable

7.3 Ownership Rule

Nightwatch must compose existing authorities rather than copy them. It may ask Sounding Line what evidence is required, ask Project Trim what context can be reused, ask Fairlead whether a GitHub query is affordable, ask Breakwater to supervise an operational process, and send state to Bridgewatch. It must not fork those contracts into Nightwatch-only alternatives.

8. Canonical Nightwatch Architecture

Nightwatch contains eight logical components:

1. Night Plan Validator - validates scope, queue, budgets, deadlines, and owner authorization.
2. Watch Dispatcher - chooses the next eligible objective.
3. Objective Controller - manages one objective's attempts and state transitions.
4. Progress Evaluator - decides whether recent work materially advanced the plan.
5. Loop Guard - fingerprints failures and enforces iteration limits.
6. Budget Broker - accounts for time, mutations, PRs, validations, and available cost signals.

7. Persistent Watch Ledger - records restart-safe state and evidence pointers.
8. Bridgewatch Projection - exposes read-only operator status and morning reports.

The architecture uses two control levels:

- Watch level: manages the whole unattended session.
- Objective level: manages one bounded task lane.

An objective-level circuit breaker normally returns control to the Watch Dispatcher. Only global exhaustion, security failure, repository-integrity failure, or a terminal queue stops the whole watch.

9. Night Plan and Work Queue

9.1 Required Night Plan Fields

Every unattended run must declare:

- Session ID.
- Repository and default branch.
- Start time and wake-up deadline.
- Global runtime and cost policy.
- Ordered objective list.
- Scope allowlist for each objective.
- Objective budgets.
- Allowed mutation types.
- Fallback tasks.
- Owner-required escalation conditions.
- Morning-report destination.

9.2 Objective Classes

Class	Meaning
PRIMARY	Core deliverable expected during the night
SECONDARY	Useful work after or between primary objectives
FALLBACK	Safe work used when primary lanes are parked or waiting
HOUSEKEEPING	Bounded cleanup, reconciliation, or archival work
REPORT_ONLY	Read-only audit or morning synthesis

9.3 Queue Scheduling

The dispatcher selects the highest-priority objective that is:

- Authorized.
- Not complete.
- Not parked for the remainder of the watch.
- Within budget.
- Not waiting on an unresolved owner decision.

- Safe against current repository state.

If a primary objective waits on CI, the dispatcher may execute an independent objective. It must return when the waiting condition changes without reconstructing the entire context.

9.4 Work-Conserving Rule

Nightwatch should avoid idle time while authorized independent work exists. Parking one objective must release capacity to the remaining queue. This is the mechanism that reinforces unattended work rather than ending it.

10. Global Night Budget

The global budget governs the whole unattended session.

Required dimensions:

- Wall-clock deadline.
- Maximum authorized work duration.
- Paid-overage policy.
- Maximum total pull requests.
- Maximum total GitHub mutations.
- Maximum simultaneous active objectives.
- Maximum repository scope.
- Maximum aggregate full validations.

Global exhaustion stops the watch after a checkpoint. Objective exhaustion parks only the objective.

10.1 Global Budget Precedence

The most restrictive applicable boundary wins. No prompt phrase, task priority, or local diagnosis may override a global ceiling.

10.2 Telemetry-Unavailable Rule

If exact credit telemetry is unavailable, Nightwatch must still enforce wall-clock, mutation, attempt, validation, and no-progress limits. "Quota unknown" never means "quota unlimited."

11. Objective Budget

Each objective receives an independent budget.

Recommended default fields:

```
{
  "objectiveId": "browser-contract-closure",
  "priority": "PRIMARY",
  "wallClockLimitMinutes": 90,
  "maxNewPullRequests": 5,
  "maxActiveCandidates": 1,
  "maxCandidateRefreezes": 2,
  "maxPrerequisitePullRequests": 2,
  "maxRepeatedFailureSignatures": 2,
  "maxProtectedMainRestarts": 2,
  "maxFullValidationRuns": 2,
  "noMaterialProgressMinutes": 45,
  "maxScopeGrowthRatio": 2.0
}
```

Budgets may be tightened by the Night Plan. Expansion requires an explicit owner-approved lease amendment and cannot occur merely because the owner is unavailable.

12. Objective State Machine

Canonical objective states:

```
PLANNED
-> PREFLIGHT
-> ACTIVE
-> WAITING_EXTERNAL
-> COMPLETE
-> PARKED_LOOP_GUARD
-> PARKED_BUDGET
-> PARKED_SCOPE
-> PARKED_OWNER_REQUIRED
-> ABORTED_INTEGRITY
```

12.1 Transition Rules

- PLANNED -> PREFLIGHT: required scope and authority route are inspected.
- PREFLIGHT -> ACTIVE: the objective has an eligible bounded execution path.
- ACTIVE -> WAITING_EXTERNAL: an external check is pending and independent work may proceed.
- ACTIVE -> COMPLETE: explicit completion predicates are satisfied.
- ACTIVE -> PARKED_LOOP_GUARD: repeated semantic failures reach the limit.
- ACTIVE -> PARKED_BUDGET: time, mutation, validation, or PR budget is exhausted.
- ACTIVE -> PARKED_SCOPE: path or diff growth exceeds authorization.
- ACTIVE -> PARKED_OWNER_REQUIRED: new authority or scope requires owner choice.
- ANY -> ABORTED_INTEGRITY: continuing could damage repository or security integrity.

Parking is a valid controlled outcome, not an agent failure to obey "continue."

13. Material Progress Model

13.1 Material Progress

Examples:

- Requested deliverable merged.
- Required prerequisite merged.
- Known blocker removed.
- Qualification advanced to a new authoritative state.
- Open or superseded PR count reduced.
- Root cause identified with new evidence.
- Objective moved closer to an explicit completion predicate.

13.2 Non-Progress

- Closing and recreating an equivalent candidate.
- Repeating unchanged tests without new evidence.

- Fetching unchanged logs repeatedly.
- Regenerating context already captured in the ledger.
- Renaming branches or PRs.
- Creating a successor with substantially the same effective diff and failure route.
- Producing more narrative about why the next attempt should work.

13.3 No-Progress Timer

Long-running commands may publish heartbeats, but a heartbeat proves liveness, not material progress. The default objective parks after 45 minutes without material progress unless its Night Plan contains a justified longer deterministic operation.

14. Iterative-Loop Detection

Nightwatch detects loops across:

- Repeated failure fingerprints.
- Equivalent candidate diff families.
- Successor PR lineage.
- Repeated protected-main reconciliation.
- Repeated full validation with no input change.
- Alternating failure patterns that return to the same state.
- Scope growth caused by absorbing each discovered prerequisite.

14.1 Cycle Rule

A cycle exists when the system returns to a previously recorded objective state without a meaningful reduction in unresolved blockers or an authoritative increase in evidence.

14.2 Alternating Loops

The detector must recognize A -> B -> A -> B patterns, not only identical consecutive failures.

14.3 Counter Reset Prohibition

The agent cannot reset loop counters by:

- Creating a new branch.
 - Creating a new PR number.
 - Renaming an objective.
 - Rewording an error.
 - Closing and reopening a candidate.
 - Starting a new chat for the same objective lineage.
-

15. Failure Fingerprints

A normalized failure fingerprint includes:

- Objective family.

- Authority mode.
- Failed workflow and job.
- Failed step.
- Stable error code or semantic class.
- Changed-path class.
- Candidate lineage.
- Protected-base relationship.
- Proposed repair category.

Example:

```
browser-contract-closure  
normal-candidate  
protected-binding  
locate-sealed-explicit-authority  
missing-qualification-envelope
```

15.1 Default Response

- First occurrence: diagnose and permit one bounded correction.
- Second equivalent occurrence: park the objective.
- Same failure after a repair claimed to address it: park immediately unless new evidence proves a distinct cause.

Fingerprints must be semantic. Different wording from GitHub, PowerShell, Node, or a wrapper must not create false novelty.

16. Park, Pivot, and Resume

16.1 Park Procedure

When parking an objective, Nightwatch must:

1. Stop new mutations for the objective.
2. Preserve current branch, commit, diff, and PR.
3. Record validation and authority evidence already completed.
4. Record the exact failure fingerprint.
5. Record budget use and attempt counts.
6. Identify the minimum safe next action.
7. Mark whether another queued objective could independently remove the blocker.
8. Return control to the Watch Dispatcher.

16.2 Pivot Procedure

The dispatcher selects the next eligible objective without losing the parked context. The remaining global budget stays available.

16.3 Resume Procedure

A parked objective may resume only when:

- The owner authorizes additional budget; or
- A separate completed objective independently removed its blocker; and

- The resume does not reset prior attempts or failure counters; and
- The Night Plan still has global budget.

An objective parked for repeated authority failure may not resume merely because main changed again.

17. Candidate and Pull-Request Lineage Control

17.1 One Active Candidate

Each deliverable may have at most one active product candidate. Prerequisite work may use separate PRs where anti-self-authorization requires it, but the product candidate remains one tracked lineage.

17.2 Draft-Until-Eligible

Where repository policy permits, the product candidate remains draft until authority prerequisites are present. It should not repeatedly become a new frozen PR before preflight proves an eligible qualification route.

17.3 Successor Limit

No more than two candidate successors are allowed by default. Each successor must record:

- Predecessor PR.
- Reason the predecessor is terminal.
- Effective diff relationship.
- New base and head.
- Remaining attempt budget.

17.4 Preflight Before Publication

Before creating or refreezing a candidate:

1. Construct the candidate locally.
2. Classify exact scope.
3. Determine authority mode.
4. Verify prerequisites exist on protected main.
5. Verify the qualification path can produce the required envelope.
6. Verify the objective has remaining attempt budget.
7. Only then create or update the candidate PR.

17.5 Supersession Hygiene

When a successor is valid:

- Close the predecessor.
 - Link predecessor and successor.
 - Preserve the terminal reason.
 - Remove obsolete active status from the ledger.
 - Schedule branch cleanup through ordinary repository policy.
-

18. Scope-Growth Control

Every objective begins with a path allowlist and a baseline effective-diff shape.

Default behavior:

Growth	Action
Up to 125% of initial path count	Continue and record
Above 125%	Explain and classify
Above 150%	Freeze product work and split prerequisites
Above 200% or more than 10 additional files	Park objective
New unrelated project/governance domain	Owner approval required

The reviewed product candidate grew from approximately 8-9 files to 32 files. Nightwatch would have parked it before this cross-domain expansion.

18.1 Discovery Is Not Authorization

Finding an unrelated defect permits recording it. It does not automatically authorize implementation. New work must fit the objective allowlist, prerequisite budget, time budget, and mutation lease.

19. Credit and Inference Efficiency

Nightwatch consumes Project Trim contracts for:

- Context capsules.
- Evidence reuse.
- Targeted context expansion.
- Cost estimation.
- Duplicate-read prevention.
- Validation reuse where authority permits.
- Compact state handoff between objectives.

19.1 Scheduling Objective

Nightwatch seeks to maximize:

Useful completed work per authorized unit of time and inference across the complete Night Plan.

It does not maximize effort spent on the current objective.

19.2 Repeat Decision

Before repeating an expensive action, Nightwatch evaluates:

- Whether inputs changed.
- Whether the previous result remains valid.
- Probability that the repeat changes the outcome.
- Remaining objective and global budget.
- Whether another task offers greater expected value.

19.3 Paid Overage

Paid overage is an explicit Night Plan field. If denied or unspecified, Nightwatch must stop before entering it. If exact telemetry is unavailable, conservative proxy limits apply.

20. Execution Leases and Mutation Governance

20.1 Execution Lease

Each active objective receives a lease containing:

- Session and objective IDs.
- Repository and branch scope.
- Path allowlist.
- Expiration.
- Remaining mutation count.
- Remaining PR and refreeze count.
- Allowed authority mode.
- Paid-overage policy.

20.2 Lease Enforcement

Mutation helpers and Nightwatch-aware workflows must reject:

- Expired leases.
- Out-of-scope branches or paths.
- Excess PR generation.
- Budget-reset attempts.
- Objective IDs without a parent Night Plan.

20.3 Interactive and Unattended Modes

Interactive work may receive broader owner steering. Unattended work must load the Night Plan and enforce its leases. This is not a prohibition on unattended desktop use; it is the contract that makes such use safe enough to strengthen.

20.4 Unrestricted Credential Risk

Strong enforcement should route unattended GitHub mutations through a budget-aware broker or constrained credential. Repository checks alone can block merges but cannot prevent unlimited PR creation. Implementation must document the enforcement level honestly.

21. Persistent Ledger and Restart Recovery

21.1 Ledger Fields

The Watch Ledger records:

- Session identity and owner authorization.
- Start, deadline, and global budget.

- Current protected-main SHA.
- Ordered objective queue.
- Active, complete, parked, and waiting objectives.
- Current branches, commits, and PRs.
- Candidate lineage and predecessors.
- Failure fingerprints.
- Validation and authority receipts.
- Last material-progress timestamp.
- Scope and mutation counters.
- Next safe action.

21.2 Restart Safety

On restart, Nightwatch must:

1. Read the ledger.
2. Fetch fresh repository state.
3. Reconcile recorded objects without resetting counters.
4. Retire stale leases.
5. Resume the dispatcher at the next eligible state.

Laptop reboot, session compaction, or a new chat must not erase objective history.

21.3 Tamper Evidence

Accepted ledgers should use stable schemas and integrity digests. Manual correction must be recorded as a new event rather than rewriting history silently.

22. Prompt and Mandate Safety

22.1 Dangerous Mandates

Nightwatch validates prompts containing phrases such as:

- Run until completion.
- Do not stop.
- Run all night.
- Retry until successful.
- Continue through every blocker.
- I will be unavailable.
- Do not stop merely to conserve quota.

These phrases are allowed only when a Night Plan provides explicit global and objective limits.

22.2 Contradiction Rule

If one instruction prioritizes conservation and another grants unlimited persistence, Nightwatch rejects the mandate and reports the conflict before starting.

22.3 Precedence

Nightwatch precedence is:

1. Security and repository integrity.
2. Global Night Plan limits.
3. Objective limits and circuit breakers.
4. Sounding Line authority.
5. Objective completion language.
6. Optional cleanup and reporting preferences.

Completion language can never outrank budgets or integrity.

23. Bridgework Command Center

Bridgework should expose a Nightwatch panel containing:

- Session start and wake-up deadline.
- Current global budget state.
- Active objective.
- Queue order.
- Completed and parked objectives.
- Runtime per objective.
- PRs created, merged, closed, and active.
- Candidate generations.
- Failure fingerprints.
- Scope-growth alarms.
- Time since material progress.
- Current CI and authority state.
- Paid-overage policy.
- Morning-report readiness.

Recommended status language:

- ON_WATCH
- WAITING_EXTERNAL
- PARKING_OBJECTIVE
- PIVOTING
- GLOBAL_BUDGET_WARNING
- WATCH_COMPLETE
- WATCH_STOPPED_INTEGRITY

Recommended severity colors:

- Green: under 50% of local budget.

- Yellow: 50-80%.
 - Orange: 80-100%.
 - Red: objective parked or global breaker active.
 - Black: mutation attempted after lease expiration.
-

24. Alerts, Escalation, and Morning Handoff

24.1 Tier 1 Alert Conditions

- Same semantic failure twice.
- Third candidate successor.
- More than five PRs in one lineage.
- Scope exceeds 200%.
- No material progress for the objective limit.
- Paid overage would begin without authorization.
- Agent attempts to extend or reset its own budget.
- Global budget is exhausted.

Objective alerts normally park the lane before notifying. They do not wait for a sleeping owner while continuing the same work.

24.2 Morning Report

The report must contain:

- Starting and ending main SHAs.
- Objectives completed.
- Objectives parked and why.
- PRs merged, closed, and left open.
- Material repairs made.
- Tests and authority results.
- Runtime and budget usage.
- Paid-overage status.
- Exact continuation actions.

The report must distinguish prerequisite progress from completion of the requested deliverable.

25. Security, Privacy, and Public-Repository Safety

- Nightwatch must not store credentials, tokens, session secrets, or private chat content in repository ledgers.
- Public incident records must minimize personal billing details unless the owner explicitly authorizes them.
- Cost records should prefer bounded operational facts such as "declared quota exhausted" and "paid overage began."

- GitHub mutation brokers must use least privilege.
- Ledger events must redact private local paths where disclosure is unnecessary.
- Bridgewatch projections must expose operational status without leaking credentials or private content.

26. Failure and Fallback Behavior

Failure	Required behavior
Exact credit telemetry unavailable	Enforce time, mutation, PR, validation, and no-progress limits
GitHub API quota low	Use Fairlead-bounded queries, local evidence, and cached results
Protected main advances	Reconcile within restart budget; park after limit
Same authority failure repeats	Park objective and pivot
CI pending	Mark waiting; work an independent objective
CI flakes once	Permit one bounded retry
CI flakes repeatedly	Fingerprint and park
Scope expands	Split prerequisite or park according to thresholds
Laptop restarts	Recover ledger without resetting counters
All objectives parked	Produce morning report and end watch
Security or integrity uncertainty	Stop global watch fail closed

Nightwatch must never interpret missing telemetry, missing authority, or missing owner presence as permission for unlimited continuation.

27. Testing and Acceptance Matrix

Area	Required proof
Night Plan validation	Invalid, conflicting, or unbounded mandates fail before execution
Queue scheduling	Blocked primary pivots to eligible secondary/fallback work
Objective budgets	Each hard limit parks the objective deterministically
Global budgets	Global exhaustion checkpoints and ends the watch
Failure fingerprints	Equivalent wording maps to one semantic failure
Alternating loops	A-B-A-B cycles are detected
Candidate lineage	Branch/PR renaming cannot reset counters
One active candidate	Duplicate live product candidates are rejected
Scope growth	125%, 150%, and 200% boundaries behave as specified
Material progress	Activity without objective advancement does not reset timer
Waiting CI	Independent objectives continue without duplicating context
Restart safety	Counters and states survive restart
Paid overage	Unauthorized transition is prevented
Bridgewatch	Projection matches authoritative ledger

Area	Required proof
Morning report	Completed, parked, and partial outcomes are distinct
Incident replay	The #302-#334 scenario parks before excessive churn and continues other work

27.1 Founding Incident Replay Requirement

The replay must model:

- Browser product candidate creation.
- Mixed Tideglass fixture failure.
- Shipwright fixture and engine prerequisites.
- Browser partition normalization.
- Governed Shipwright runner admission.
- P34 ledger admission.
- Repeated missing explicit authority.
- Scope growth.
- Candidate successor churn.

Acceptance requires that the browser lane parks no later than the earliest of:

- Five generated PRs.
- Two hours of objective time.
- The second equivalent explicit-authority failure.
- 200% scope growth.

The dispatcher must then continue with another authorized objective.

27.2 Adversarial Cases

- Error text changes while semantic cause remains.
- Agent creates a new objective ID to evade counters.
- Agent closes and recreates branches.
- Main advances continuously.
- CI never finishes.
- Test flakes alternate with authority failures.
- Credit telemetry disappears.
- Supervisor restarts.
- Worktree survives but remote branch disappears.
- All fallback objectives are exhausted.

28. Implementation Phases

Phase 0 - Preserve the Watch

- Record the founding Tier 1 incident.
- Preserve the #302-#334 lineage as replay evidence.

- Record prompt contradiction and impact.
- Freeze the final reviewed state.
- Establish Nightwatch ownership proposal.

Phase 1 - Set the Night Plan

- Implement Night Plan schema and validator.
- Define global and objective budgets.
- Implement prompt contradiction checks.
- Define objective state transitions.
- Register Nightwatch governance and cross-project boundaries.

Phase 2 - Keep the Ledger

- Implement persistent Watch Ledger.
- Implement restart and compaction recovery.
- Track candidate lineage, failures, scope, and progress.
- Integrate Project Trim context capsules.

Phase 3 - Guard the Lanes

- Implement failure fingerprints.
- Implement loop and alternating-cycle detection.
- Implement scope-growth tripwires.
- Implement material-progress timers.
- Implement park, pivot, and resume.

Phase 4 - Run the Watch

- Implement Watch Dispatcher and Objective Controller.
- Implement execution leases and mutation accounting.
- Integrate Sounding Line state.
- Integrate Fairlead GitHub quota behavior.
- Integrate Breakwater runtime supervision.

Phase 5 - Raise the Command Center

- Add Nightwatch state to Bridgework.
- Add alerts, budgets, queue, and progress displays.
- Add morning reports and continuation controls.
- Add incident-history navigation.

Phase 6 - Prove the Dawn

- Replay the founding incident.
- Run adversarial scenarios.
- Prove restart safety.
- Conduct a supervised synthetic overnight session.

- Demonstrate useful work continues after a lane parks.
 - Activate Nightwatch as the required unattended operating contract.
-

29. Final Acceptance Criteria

Nightwatch v1.0 is accepted only when:

1. Unattended work remains supported and can span an authorized night.
2. Every run has a validated Night Plan.
3. Global and objective budgets are independently enforced.
4. A parked objective does not unnecessarily end the session.
5. Equivalent repeated failures are detected semantically.
6. No objective exceeds its PR, refreeze, prerequisite, or scope limits.
7. One active product candidate exists per deliverable.
8. Counters cannot be reset by new branches, PRs, chats, or wording.
9. Missing credit telemetry cannot produce unlimited work.
10. Unauthorized paid overage is prevented by available enforcement boundaries.
11. The ledger survives restart and reflects remote reality.
12. Bridgewatch displays authoritative session state.
13. The morning report distinguishes completion, partial progress, and parked blockers.
14. The founding incident replay parks the lane before excessive churn.
15. After parking that lane, another authorized objective completes successfully.

The governing success condition is not "Codex never fails." It is:

No unattended objective may silently continue beyond its authorized time, attempt, scope, PR, validation, or cost boundaries, and no single pathological objective may consume the useful remainder of an authorized night.

30. Governance and Change Control

30.1 Authority

This document defines the Nightwatch governing baseline. Implementation records may refine mechanics but may not weaken the non-negotiable principles without a versioned governing amendment.

30.2 Threshold Changes

Default thresholds may be tuned from measured evidence. Any increase must document:

- The workload class requiring it.
- Expected benefit.
- New maximum exposure.
- Replay and adversarial evidence.
- Owner approval.

30.3 Incident Amendments

New runaway or near-miss incidents must append evidence and tests. History must not be rewritten to make the system appear more successful than it was.

30.4 Cross-Project Amendments

Changes affecting Sounding Line authority, Project Trim evidence reuse, Breakwater operations, Fairlead quota control, Bridgewatch projection, or Deepwater ownership must follow those projects' existing governance and acceptance paths.

30.5 No Self-Authorization

Nightwatch cannot modify its own budget enforcement and use the modified behavior to qualify that same change. Anti-self-authorization boundaries remain mandatory.

31. Glossary

Term	Definition
Night Plan	Authorized manifest for one unattended session
Watch	The complete unattended session
Objective	One bounded task lane inside a watch
Global budget	Limits applying to the whole watch
Objective budget	Limits applying to one objective
Material progress	Evidence that an objective measurably advanced
Failure fingerprint	Semantic identity for a failure across changing surface details
Park	Preserve and suspend a pathological or exhausted objective
Pivot	Move capacity to another authorized objective
Resume	Continue a parked objective without resetting history
Execution lease	Bounded authorization for an objective's operations
Watch Ledger	Persistent authoritative record of session state
Candidate lineage	Related branches and PRs pursuing the same deliverable
Morning handoff	Final report for the returning owner

32. Appendix A - Default Thresholds

Metric	Warning	Default action limit
Objective wall time	60 minutes	Park at 90 minutes
No material progress	20 minutes	Park at 45 minutes
Same semantic failure	First repeat	Park at second occurrence
Candidate successors	1	Park before third
Prerequisite PRs	1	Park before third
New PRs in one lineage	3	Park at 5
Concurrent product candidates	N/A	Reject above 1

Metric	Warning	Default action limit
Protected-main restarts	1	Park after 2
Full qualification runs	1	Park after 2 without progress
Scope growth	125%	Park at 200% or +10 files
Global budget	50%	Warn at 80%; stop at 100%

Thresholds are defaults, not universal estimates. The Night Plan may assign smaller limits. Larger limits require owner-approved evidence.

33. Appendix B - Founding Incident Evidence

B.1 Merged Repairs

PR	Outcome
#305	Mixed Tideglass browser fixtures partitioned
#311	Shipwright browser fixture isolated
#315	Shipwright browser engine registered
#318	Browser-selection partitions normalized
#330	Governed Shipwright runner admitted
#333	Generated P34 browser ledgers admitted

B.2 Terminal Reviewed Candidate

Field	Value
PR	#334
Head	511570d723aad5ba17a9a42f2eaf6c23a9611bce
Base	d7e3f2f873139e3f7353bf288124f50543ce2aac
Files	32
Additions	1,671
Deletions	1,089
Protected binding	Failed
Failure step	Locate sealed explicit authority

B.3 Counterfactual Nightwatch Outcome

Nightwatch would have:

1. Allowed initial focused diagnosis.
2. Allowed a bounded prerequisite repair.
3. Recognized repeated authority and candidate-lineage failures.
4. Parked the product lane before excessive PR generation or scope growth.
5. Continued with Tideglass cleanup and supersession work.
6. Produced a morning report with preserved state and unused global capacity.

34. Appendix C - Example Night Plan

```
{
  "schemaVersion": "1.0.0",
  "sessionId": "nightwatch-2026-08-20-home",
  "repository": "Kgray44/treasurehuntSoT",
  "baseBranch": "main",
  "mode": "unattended",
  "wakeDeadline": "2026-08-20T07:00:00-04:00",
  "globalBudget": {
    "wallClockMinutes": 420,
    "maxNewPullRequests": 8,
    "maxFullQualificationRuns": 6,
    "paidOverage": false
  },
  "objectives": [
    {
      "id": "browser-contract-closure",
      "class": "PRIMARY",
      "priority": 1,
      "wallClockMinutes": 90,
      "maxNewPullRequests": 5,
      "maxCandidateRefreezes": 2,
      "maxRepeatedFailureSignatures": 2,
      "noMaterialProgressMinutes": 45
    },
    {
      "id": "tideglass-fixture-closure",
      "class": "PRIMARY",
      "priority": 2,
      "wallClockMinutes": 75,
      "maxNewPullRequests": 3
    },
    {
      "id": "superseded-pr-cleanup",
      "class": "HOUSEKEEPING",
      "priority": 3,
      "wallClockMinutes": 45,
      "maxNewPullRequests": 0
    },
    {
      "id": "morning-repository-capsule",
      "class": "REPORT_ONLY",
      "priority": 99,
      "wallClockMinutes": 20
    }
  ]
}
```

35. Appendix D - Example Morning Report

PROJECT NIGHTWATCH - MORNING HANDOFF

Watch: nightwatch-2026-08-20-home
Runtime: 6h 41m of 7h authorized
Paid overage: not entered
Starting main: <sha>
Ending main: <sha>

COMPLETE

- Tideglass fixture closure: merged PR #...
- Superseded PR cleanup: 11 closed, 0 ambiguous
- Repository capsule: published

PARKED

- Browser product contracts
Reason: repeated authority.normal-candidate.missing-explicit-envelope
Attempts: 2/2
PRs created: 3/5
Last candidate: #...
Preserved head: <sha>
Minimum next action: inspect candidate authority dispatch contract

GLOBAL BUDGET

- Runtime used: 95%
- PR budget used: 50%
- Full qualifications: 3/6
- Remaining objectives: none eligible

No security bypass, protection change, or unapproved paid usage occurred.

End of Project Nightwatch Governing Document v1.0